

DitroniX
Home Automation | Internet of Things | Energy Monitoring

(C)2018-2026 Dave Williams | DitroniX | G8PUO

For further information and code examples, please see our code and GitHub WIKI pages. ditronix.net | [GitHub.com/DitroniX](https://github.com/DitroniX) | [Hackster.io/DitroniX](https://hackster.io/DitroniX)



IPEM SIX ESP32-C5-WROOM-1-N8R8 ATM90E32AS 1.2607.102			
DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG1)			
Sheet:	1/4	Drwg/Chk: Dave Williams	09/07/2026 03:31

Bank A

IPEM SIX is an IoT Power Energy Monitor.
IPEM SIX is an ESP32-C5 variant of the IPEM and EPEM series.



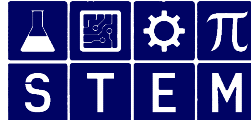
Production

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Enclosure VG-GR34 90x71x57
Main PCB 86 x 66.8

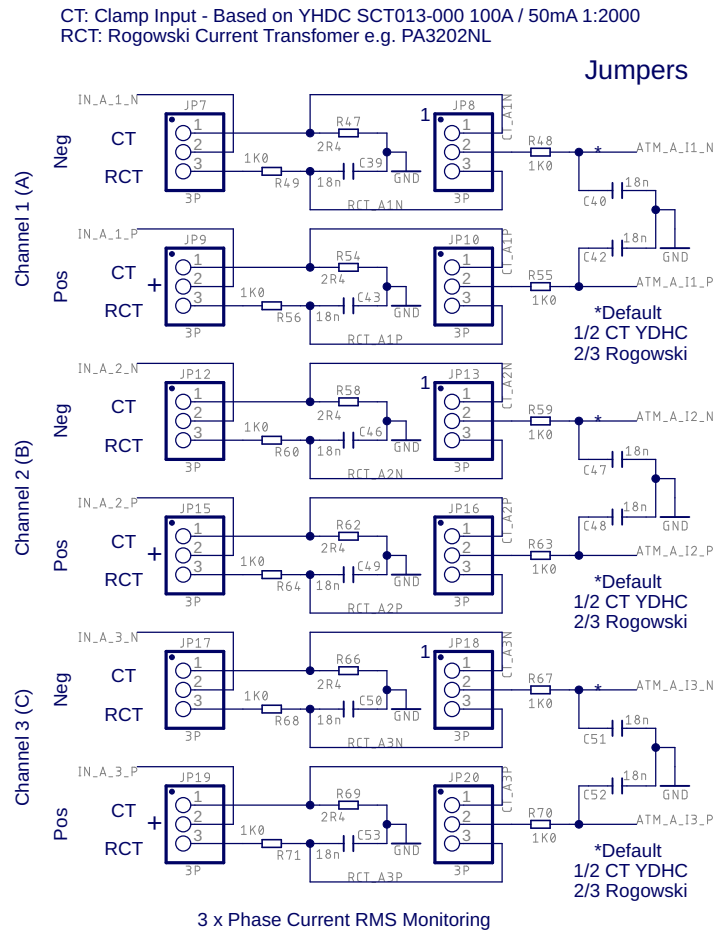
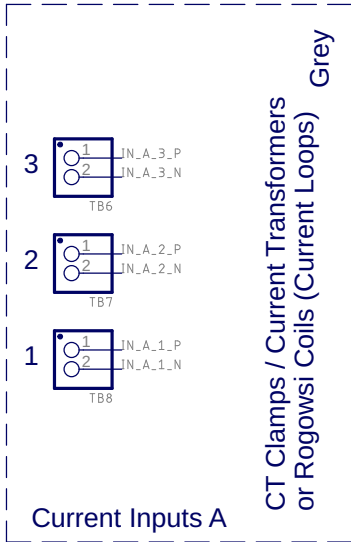
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Components optimised for BOM

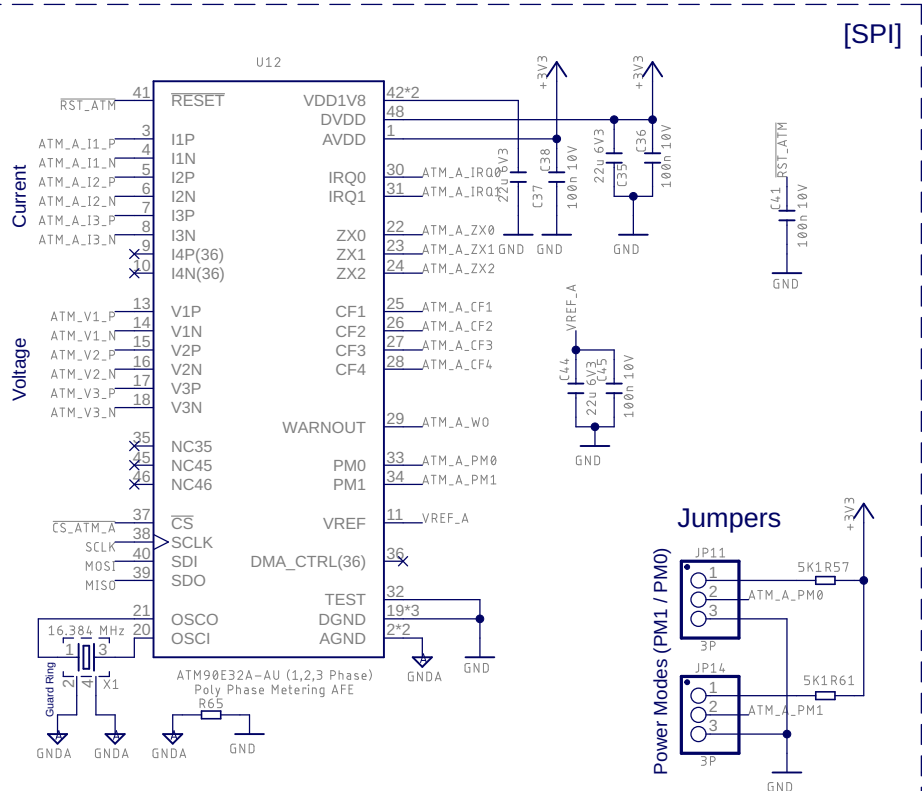
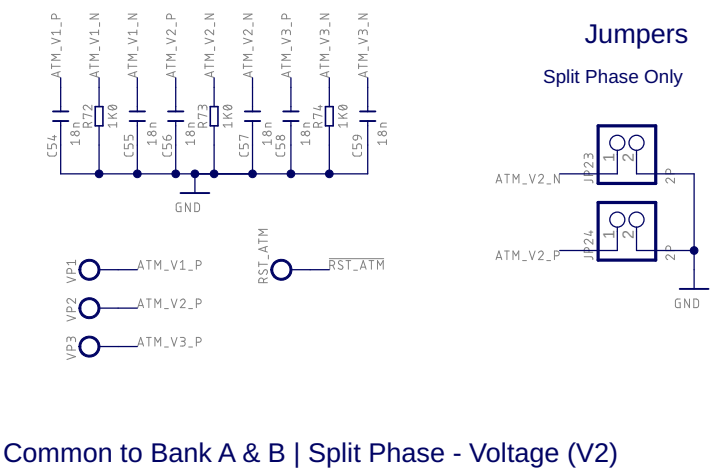


ESP32-C5 ATM90E32AS SDK
ditronix.net
Ditronix Ltd – UK

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DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG3)			
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Current Sensor Inputs Select | CT Clamp | Rogowski



AC Voltage Sense is Common to both Bank A and B

Power Modes (PM1/PM0)

Normal (N) *	1+0 ON
Partial Measure	Only 0 ON
Detection (D)	Only 1 ON
Idle (I)	ALL OFF

ATM_WO - Warning Out
High = Error

CF1 - Active (Red)
CF2 - ReActive (Green)
CF3 - Fundamental (Blue)
CF4 - Harmonic (Amber/Ylw)
Pulse Outputs Status LEDs

Defaults

WO - L
ZXx - HL
IRQx - L
CFx - L

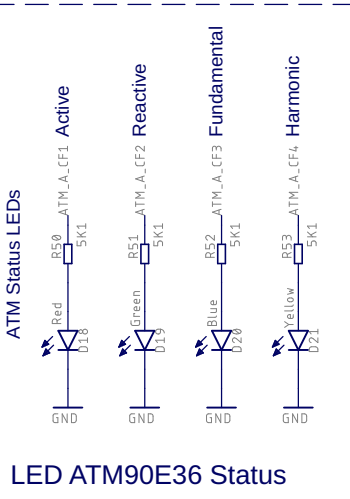
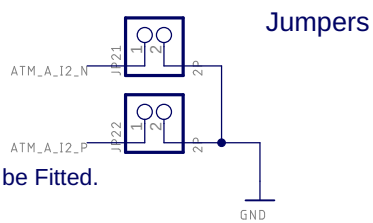
Mains AC Power Energy Monitor | ATM90E32AS Bank A

SPLIT Phase ONLY = ON (e.g. USA)

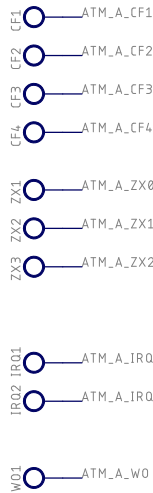
Only Inputs 1 & 3 are used for SPLIT Phase

If Split Phase ALL Current and Voltage Jumpers to be Fitted.

Split Phase - Current (I2)



LED ATM90E36 Status



Test Points

Bank B

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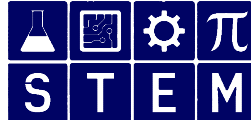
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Components optimised for BOM



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DUAL-ATM90E32-ESP32-C5-IOT-POWER-ENERGY-MONITOR-BOARD-(PG4)			
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